

Technical Document #973: Computer Vision - Comprehensive Overview

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Image generation creates new images from random noise or text descriptions. Diffusion models like Stable Diffusion produce photorealistic images.

Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Face recognition identifies or verifies faces in images. DeepFace and FaceNet achieve near-human accuracy on face verification tasks.

Object detection identifies and localizes objects in images. YOLO and Faster R-CNN are popular real-time object detection architectures.

Video understanding analyzes temporal dynamics in video sequences. 3D CNNs and transformer-based models capture spatiotemporal features.

Image super-resolution reconstructs high-resolution images from low-resolution inputs. GAN-based methods produce photorealistic upscaled images.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Image generation creates new images from random noise or text descriptions.
- Semantic segmentation classifies each pixel in an image into a category.
- Visual question answering (VQA) answers questions about images.